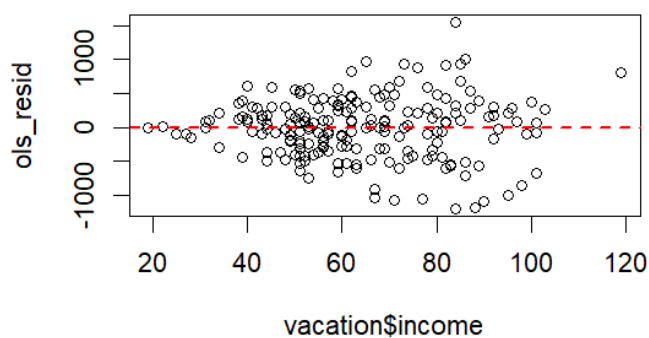


8.16

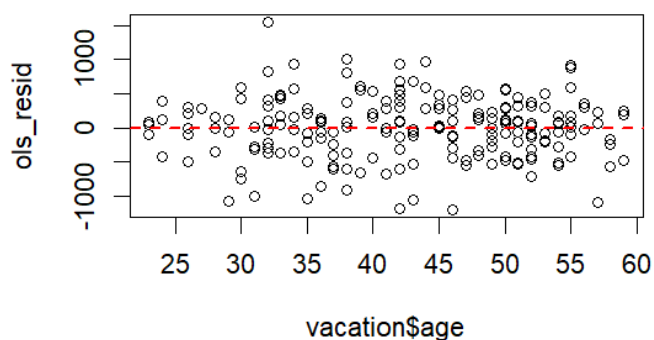
(a)

We are 95% confident that, holding income and age constant, adding one more child to the household changes the miles traveled for a vacation by between -135.3298 and -28.32302 miles per year.

(b)



This classic "funnel" or "megaphone" shape is strong visual evidence of **heteroskedasticity**. It indicates that the variance of the error term is not constant; it increases as a household's income increases.



There is no obvious visual evidence of heteroskedasticity related to the age variable.

(c)

$$H_0: \sigma_{low}^2 = \sigma_{high}^2$$

$$H_1: \sigma_{low}^2 < \sigma_{high}^2$$

--- Goldfeld-Quandt Test Results ---

- SSE (Low Income): 8750040
- SSE (High Income): 27160654
- F-Statistic: 3.104061
- Critical Value (5%): 1.428617

Because the test statistic is greater than the critical value, we reject the null hypothesis at the 5% significance level. There is sufficient evidence to conclude that heteroskedasticity is present and the variance depends on income.

(d)

--- Confidence Intervals for kids ---

1. Original OLS CI : [-135.3298, -28.32302]

2. Robust CI : [-139.32297, -24.32986]

Comparing the two results, the 95% robust confidence interval for the kids coefficient is wider than the standard OLS confidence interval calculated in part (a). In parts (b) and (c), we found strong evidence of heteroskedasticity related to income. When heteroskedasticity is present, conventional OLS standard errors are biased and often underestimate the true variance. This makes the standard confidence intervals artificially narrow. By using heteroskedasticity-robust standard errors in part (d), we correct for this bias. The resulting robust interval is a more reliable and accurate estimate of the true effect of having one more child on vacation miles traveled.

(e.)

--- 95% Confidence Intervals for 'KIDS' ---

1. OLS Usual (Part a): [-135.3298, -28.32302]
2. OLS Robust (Part d): [-139.32297, -24.32986]
3. GLS Conventional (Part e): [-119.8945, -33.71808]
4. GLS Robust (Part e): [-121.41339, -32.19919]

The conventional GLS interval is the narrowest because it explicitly corrects for the heteroskedasticity, resulting in a more efficient estimator than OLS. The robust GLS interval is slightly wider than the conventional GLS interval to account for potential misspecification of the exact variance function, but it generally remains narrower and more efficient than the purely robust OLS interval from part (d)